

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version	Revision Date:	Date of last issue: 13.11.2019	Print Date:
2.6	07.07.2020	Date of first issue: 05.11.2014	07.07.2020

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name : Klübersynth GEM 4-150 N

Article-No. : 012233

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Sub-
stance/Mixture : Lubricating oil

Recommended restrictions
on use : Restricted to professional users.

1.3 Details of the supplier of the safety data sheet

Company : Klüber Lubrication München
Geisenhausenerstr. 7
81379 München
Deutschland
Tel: +49 (0) 89 7876 0
Fax: +49 (0) 89 7876 333
info@klueber.com

E-mail address of person
responsible for the SDS : mcm@klueber.com
Material Compliance Management

National contact : Klüber Lubrication Great Britain Limited
Unit 10 Pennine Business Park
Longbow Close
Huddersfield
West Yorkshire HD2 1GQ
Great Britain
Tel: +44-1422-205115
Fax: +44-1422-206073
sales@uk.klueber.com

1.4 Emergency telephone number

Emergency telephone num-
ber : +49 89 7876 700 (24 hrs)

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)
Not a hazardous substance or mixture.

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version	Revision Date:	Date of last issue: 13.11.2019	Print Date:
2.6	07.07.2020	Date of first issue: 05.11.2014	07.07.2020

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

Not a hazardous substance or mixture.

Additional Labelling

EUH210 Safety data sheet available on request.
EUH208 Contains amines, C12-14-tert-alkyl. May produce an allergic reaction.

2.3 Other hazards

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Chemical nature : Synthetic hydrocarbon oil
ester oil

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number	Classification	Concentration limits M-Factor Notes	Concentration (% w/w)
amines, C12-14-tert-alkyl	701-175-2 01-2119456798-18-XXXX	Acute Tox.4; H302 Acute Tox.2; H330 Acute Tox.3; H311 Skin Corr.1B; H314 Eye Dam.1; H318 Skin Sens.1A; H317 STOT SE3; H335 Aquatic Acute1; H400 Aquatic Chronic1; H410	M-Factor: 1/1	>= 0.025 - < 0.1

For explanation of abbreviations see section 16.

SECTION 4: First aid measures

4.1 Description of first aid measures

If inhaled : Remove person to fresh air. If signs/symptoms continue, get medical attention.
Keep patient warm and at rest.

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version 2.6	Revision Date: 07.07.2020	Date of last issue: 13.11.2019 Date of first issue: 05.11.2014	Print Date: 07.07.2020
----------------	------------------------------	---	---------------------------

If breathing is irregular or stopped, administer artificial respiration.

In case of skin contact : Remove contaminated clothing. If irritation develops, get medical attention.
In case of contact, immediately flush skin with plenty of water.

In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 10 minutes.
If eye irritation persists, consult a specialist.

If swallowed : Move the victim to fresh air.
Do NOT induce vomiting.
Rinse mouth with water.

4.2 Most important symptoms and effects, both acute and delayed

Symptoms : No information available.

Risks : None known.

4.3 Indication of any immediate medical attention and special treatment needed

Treatment : No information available.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media : Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

Unsuitable extinguishing media : High volume water jet

5.2 Special hazards arising from the substance or mixture

Specific hazards during fire-fighting : Fire may cause evolution of:
Carbon oxides

5.3 Advice for firefighters

Special protective equipment for firefighters : In the event of fire, wear self-contained breathing apparatus.
Use personal protective equipment. Exposure to decomposition products may be a hazard to health.

Further information : Standard procedure for chemical fires.

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version	Revision Date:	Date of last issue: 13.11.2019	Print Date:
2.6	07.07.2020	Date of first issue: 05.11.2014	07.07.2020

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions : Evacuate personnel to safe areas.
Use personal protective equipment.
Refer to protective measures listed in sections 7 and 8.

6.2 Environmental precautions

Environmental precautions : Try to prevent the material from entering drains or water courses.
Prevent further leakage or spillage if safe to do so.
Local authorities should be advised if significant spillages cannot be contained.

6.3 Methods and material for containment and cleaning up

Methods for cleaning up : Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).

6.4 Reference to other sections

For personal protection see section 8.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advice on safe handling : For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Wash hands and face before breaks and immediately after handling the product.

Hygiene measures : Wash face, hands and any exposed skin thoroughly after handling.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : Store in original container. Keep container closed when not in use. Keep in a dry, cool and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage. Store in accordance with the particular national regulations. Keep in properly labelled containers.

7.3 Specific end use(s)

Specific use(s) : Specific instructions for handling, not required.

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version	Revision Date:	Date of last issue: 13.11.2019	Print Date:
2.6	07.07.2020	Date of first issue: 05.11.2014	07.07.2020

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Contains no substances with occupational exposure limit values.

Derived No Effect Level (DNEL) according to Regulation (EC) No. 1907/2006:

Substance name	End Use	Exposure routes	Potential health effects	Value
diisotridecyl adipate	Workers	Inhalation	Long-term systemic effects	24 mg/m3
	Workers	Skin contact	Long-term systemic effects	3.4 mg/kg bw/day
amines, C12-14-tert-alkyl	Workers	Inhalation	Long-term systemic effects	12.5 mg/m3
	Workers	Inhalation	Long-term local effects	12.1 mg/m3

Predicted No Effect Concentration (PNEC) according to Regulation (EC) No. 1907/2006:

Substance name	Environmental Compartment	Value
diisotridecyl adipate	Fresh water	0.00004 mg/l
	Intermittent use/release	1 mg/l
	Marine water	0.000004 mg/l
	Intermittent use/release	1 mg/l
	Sewage treatment plant	48 mg/l
	Fresh water sediment	40 mg/kg dry weight (d.w.)
amines, C12-14-tert-alkyl	Marine sediment	40 mg/kg dry weight (d.w.)
	Soil	1 mg/kg dry weight (d.w.)
	Fresh water	0.001 mg/l
	Marine water	0.0001 mg/l
	Intermittent use/release	0.004 mg/l
	Sewage treatment plant	0.635 mg/l
	Fresh water sediment	2.14 mg/l
	Marine sediment	0.214 mg/l
	Soil	0.428 mg/l
	Oral	4.71 mg/l

8.2 Exposure controls

Engineering measures

none

Personal protective equipment

Eye protection : Safety glasses with side-shields

Hand protection

Material : Nitrile rubber
Break through time : > 10 min
Protective index : Class 1

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version 2.6	Revision Date: 07.07.2020	Date of last issue: 13.11.2019 Date of first issue: 05.11.2014	Print Date: 07.07.2020
----------------	------------------------------	---	---------------------------

Remarks	: Wear protective gloves. The break through time depends amongst other things on the material, the thickness and the type of glove and therefore has to be measured for each case. The selected protective gloves have to satisfy the specifications of Regulation (EU) 2016/425 and the standard EN 374 derived from it.
Respiratory protection	: Not required; except in case of aerosol formation.
Filter type	: Filter type A-P
Protective measures	: The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace. Choose body protection in relation to its type, to the concentration and amount of dangerous substances, and to the specific work-place.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Appearance	: liquid
Colour	: yellow
Odour	: characteristic
Odour Threshold	: No data available
pH	: No data available
Melting point/range	: No data available
Boiling point/boiling range	: No data available
Flash point	: $\geq 200\text{ °C}$ Method: open cup
Evaporation rate	: No data available
Flammability (solid, gas)	: Not applicable
Upper explosion limit / Upper flammability limit	: No data available
Lower explosion limit / Lower	: No data available

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version 2.6	Revision Date: 07.07.2020	Date of last issue: 13.11.2019 Date of first issue: 05.11.2014	Print Date: 07.07.2020
----------------	------------------------------	---	---------------------------

flammability limit

Vapour pressure : < 0.001 hPa (20 °C)

Relative vapour density : No data available

Density : 0.86 g/cm³
(20 °C)

Bulk density : No data available

Solubility(ies)
Water solubility : insoluble

Solubility in other solvents : No data available

Partition coefficient: n-octanol/water : No data available

Auto-ignition temperature : No data available

Decomposition temperature : No data available

Viscosity
Viscosity, dynamic : No data available
Viscosity, kinematic : 150 mm²/s (40 °C)

Explosive properties : Not explosive

Oxidizing properties : No data available

9.2 Other information

Sublimation point : No data available

Self-ignition : No data available

SECTION 10: Stability and reactivity

10.1 Reactivity

No hazards to be specially mentioned.

10.2 Chemical stability

Stable under normal conditions.

10.3 Possibility of hazardous reactions

Hazardous reactions : No dangerous reaction known under conditions of normal use.

10.4 Conditions to avoid

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version 2.6	Revision Date: 07.07.2020	Date of last issue: 13.11.2019 Date of first issue: 05.11.2014	Print Date: 07.07.2020
----------------	------------------------------	---	---------------------------

Conditions to avoid : No conditions to be specially mentioned.

10.5 Incompatible materials

Materials to avoid : No materials to be especially mentioned.

10.6 Hazardous decomposition products

No decomposition if stored and applied as directed.

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Acute toxicity

Product:

Acute oral toxicity : Remarks: This information is not available.
Acute inhalation toxicity : Remarks: This information is not available.
Acute dermal toxicity : Remarks: This information is not available.

Components:

amines, C12-14-tert-alkyl:

Acute oral toxicity : LD50 (Rat): 612 mg/kg
Method: OECD Test Guideline 401

Acute inhalation toxicity : LC50 (Rat): 1.19 mg/l
Exposure time: 4 h
Test atmosphere: vapour
Method: OECD Test Guideline 403

Acute dermal toxicity : LD50 (Rat): 251 mg/kg
Method: OECD Test Guideline 402

Skin corrosion/irritation

Product:

Remarks : This information is not available.

Components:

amines, C12-14-tert-alkyl:

Assessment : Causes burns.
Result : Causes burns.

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version	Revision Date:	Date of last issue: 13.11.2019	Print Date:
2.6	07.07.2020	Date of first issue: 05.11.2014	07.07.2020

Serious eye damage/eye irritation

Product:

Remarks : This information is not available.

Components:

amines, C12-14-tert-alkyl:

Species : Rabbit
Assessment : Risk of serious damage to eyes.
Method : OECD Test Guideline 405
Result : Irreversible effects on the eye

Respiratory or skin sensitisation

Product:

Remarks : This information is not available.

Components:

amines, C12-14-tert-alkyl:

Species : Guinea pig
Assessment : The product is a skin sensitiser, sub-category 1A.
Method : OECD Test Guideline 406
Result : Probability or evidence of high skin sensitisation rate in humans

Germ cell mutagenicity

Product:

Genotoxicity in vitro : Remarks: No data available

Genotoxicity in vivo : Remarks: No data available

Components:

amines, C12-14-tert-alkyl:

Genotoxicity in vitro : Test Type: In vitro mammalian cell gene mutation test
Test system: Chinese hamster ovary cells
Method: OECD Test Guideline 476
Result: negative

Test Type: Ames test
Test system: Salmonella typhimurium
Method: OECD Test Guideline 471
Result: negative

Genotoxicity in vivo : Test Type: Chromosome aberration test in vitro
Species: Mouse
Application Route: Oral

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version 2.6	Revision Date: 07.07.2020	Date of last issue: 13.11.2019 Date of first issue: 05.11.2014	Print Date: 07.07.2020
----------------	------------------------------	---	---------------------------

Method: OECD Test Guideline 474

Result: negative

Germ cell mutagenicity- Assessment : Animal testing did not show any mutagenic effects.

Carcinogenicity

Product:

Remarks : No data available

Reproductive toxicity

Product:

Effects on fertility : Remarks: No data available

Effects on foetal development : Remarks: No data available

Components:

amines, C12-14-tert-alkyl:

Effects on foetal development : Species: Rat
Method: OECD Test Guideline 414
Result: No effects on fertility and early embryonic development were detected.

Reproductive toxicity - Assessment : Animal testing did not show any effects on fertility.
Animal testing did not show any effects on foetal development.

STOT - single exposure

Components:

amines, C12-14-tert-alkyl:

Assessment : May cause respiratory irritation.

STOT - repeated exposure

Components:

amines, C12-14-tert-alkyl:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity

Product:

Remarks : This information is not available.

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version	Revision Date:	Date of last issue: 13.11.2019	Print Date:
2.6	07.07.2020	Date of first issue: 05.11.2014	07.07.2020

Components:

amines, C12-14-tert-alkyl:

Species	:	Rat
NOAEL	:	19 mg/kg
Application Route	:	Inhalation
Test atmosphere	:	vapour
Method	:	OECD Test Guideline 412

Species	:	Rat
NOAEL	:	20 mg/kg
Application Route	:	Skin contact
Method	:	OECD Test Guideline 410

Aspiration toxicity

Product:

This information is not available.

Components:

amines, C12-14-tert-alkyl:

No aspiration toxicity classification

Further information

Product:

Remarks	:	Information given is based on data on the components and the toxicology of similar products.
---------	---	--

SECTION 12: Ecological information

12.1 Toxicity

Product:

Toxicity to fish	:	Remarks: No data available
------------------	---	----------------------------

Toxicity to daphnia and other aquatic invertebrates	:	Remarks: No data available
---	---	----------------------------

Toxicity to algae/aquatic plants	:	Remarks: No data available
----------------------------------	---	----------------------------

Toxicity to microorganisms	:	Remarks: No data available
----------------------------	---	----------------------------

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version	Revision Date:	Date of last issue: 13.11.2019	Print Date:
2.6	07.07.2020	Date of first issue: 05.11.2014	07.07.2020

Components:

amines, C12-14-tert-alkyl:

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 1.3 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 2.5 mg/l
Exposure time: 48 h

Toxicity to algae/aquatic plants : EC50 (Selenastrum capricornutum (green algae)): 0.44 mg/l
Exposure time: 72 h
Test Type: Growth inhibition
Method: OECD Test Guideline 201

M-Factor (Acute aquatic toxicity) : 1

Toxicity to fish (Chronic toxicity) : NOEC: 0.078 mg/l
Exposure time: 96 d
Species: Oncorhynchus mykiss (rainbow trout)
Method: OECD Test Guideline 210

M-Factor (Chronic aquatic toxicity) : 1

12.2 Persistence and degradability

Product:

Biodegradability : Remarks: No data available

Physico-chemical removability : Remarks: No data available

Components:

amines, C12-14-tert-alkyl:

Biodegradability : Inoculum: activated sludge
Concentration: 4 mg/l
Result: Not rapidly biodegradable
Biodegradation: 21.8 %
Exposure time: 28 d
Method: OECD Test Guideline 301D

12.3 Bioaccumulative potential

Product:

Bioaccumulation : Remarks: This mixture contains no substance considered to be persistent, bioaccumulating and toxic (PBT).
This mixture contains no substance considered to be very persistent and very bioaccumulating (vPvB).

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version	Revision Date:	Date of last issue: 13.11.2019	Print Date:
2.6	07.07.2020	Date of first issue: 05.11.2014	07.07.2020

Components:

amines, C12-14-tert-alkyl:

Partition coefficient: n-octanol/water : log Pow: 2.9 (20 °C)

12.4 Mobility in soil

Product:

Mobility : Remarks: No data available

Distribution among environmental compartments : Remarks: No data available

12.5 Results of PBT and vPvB assessment

Product:

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher..

12.6 Other adverse effects

Product:

Additional ecological information : No information on ecology is available.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product : The product should not be allowed to enter drains, water courses or the soil.

Waste codes should be assigned by the user based on the application for which the product was used.

Contaminated packaging : Packaging that is not properly emptied must be disposed of as the unused product.
Dispose of waste product or used containers according to local regulations.

The following Waste Codes are only suggestions:

Waste Code : unused product
13 02 06*, synthetic engine, gear and lubricating oils

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version 2.6	Revision Date: 07.07.2020	Date of last issue: 13.11.2019 Date of first issue: 05.11.2014	Print Date: 07.07.2020
----------------	------------------------------	---	---------------------------

uncleaned packagings
15 01 10, packaging containing residues of or contaminated
by hazardous substances

SECTION 14: Transport information

14.1 UN number

ADR	:	Not regulated as a dangerous good
IMDG	:	Not regulated as a dangerous good
IATA	:	Not regulated as a dangerous good

14.2 UN proper shipping name

ADR	:	Not regulated as a dangerous good
IMDG	:	Not regulated as a dangerous good
IATA	:	Not regulated as a dangerous good

14.3 Transport hazard class(es)

ADR	:	Not regulated as a dangerous good
IMDG	:	Not regulated as a dangerous good
IATA	:	Not regulated as a dangerous good

14.4 Packing group

ADR	:	Not regulated as a dangerous good
IMDG	:	Not regulated as a dangerous good
IATA (Cargo)	:	Not regulated as a dangerous good
IATA (Passenger)	:	Not regulated as a dangerous good

14.5 Environmental hazards

ADR	:	Not regulated as a dangerous good
IMDG	:	Not regulated as a dangerous good
IATA (Passenger)	:	Not regulated as a dangerous good
IATA (Cargo)	:	Not regulated as a dangerous good

14.6 Special precautions for user

Not applicable

14.7 Transport in bulk according to Annex II of Marpol and the IBC Code

Remarks	:	Not applicable for product as supplied.
---------	---	---

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version	Revision Date:	Date of last issue: 13.11.2019	Print Date:
2.6	07.07.2020	Date of first issue: 05.11.2014	07.07.2020

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

- | | | |
|---|---|--|
| REACH - Candidate List of Substances of Very High Concern for Authorisation (Article 59). | : | This product does not contain substances of very high concern (Regulation (EC) No 1907/2006 (REACH), Article 57). |
| REACH - List of substances subject to authorisation (Annex XIV) | : | Not applicable |
| Regulation (EC) No 1005/2009 on substances that deplete the ozone layer | : | Not applicable |
| Regulation (EU) 2019/1021 on persistent organic pollutants (recast) | : | Not applicable |
| Regulation (EC) No 649/2012 of the European Parliament and the Council concerning the export and import of dangerous chemicals | : | Not applicable |
| REACH - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, preparations and articles (Annex XVII) | : | Not applicable |
| Seveso III: Directive 2012/18/EU of the European Parliament and of the Council on the control of major-accident hazards involving dangerous substances. | | Not applicable |
| Volatile organic compounds | : | Directive 2010/75/EU of 24 November 2010 on industrial emissions (integrated pollution prevention and control)
Volatile organic compounds (VOC) content: 0.12 % |

15.2 Chemical safety assessment

This information is not available.

SECTION 16: Other information

Full text of H-Statements

- | | | |
|------|---|---|
| H302 | : | Harmful if swallowed. |
| H311 | : | Toxic in contact with skin. |
| H314 | : | Causes severe skin burns and eye damage. |
| H317 | : | May cause an allergic skin reaction. |
| H318 | : | Causes serious eye damage. |
| H330 | : | Fatal if inhaled. |
| H335 | : | May cause respiratory irritation. |
| H400 | : | Very toxic to aquatic life. |
| H410 | : | Very toxic to aquatic life with long lasting effects. |

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version	Revision Date:	Date of last issue: 13.11.2019	Print Date:
2.6	07.07.2020	Date of first issue: 05.11.2014	07.07.2020

Full text of other abbreviations

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - European Agreement concerning the International Carriage of Dangerous Goods by Road; AICS - Australian Inventory of Chemical Substances; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of Very High Concern; TCSI - Taiwan Chemical Substance Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

Further information

This safety data sheet applies only to products as originally packed and labelled. The information contained therein may not be reproduced or modified without our express written permission. Any forwarding of this document is only permitted to the extent required by law. Any further, in particular public, dissemination of the safety data sheet (e.g. as a document for download from the Internet) is not permitted without our express written consent. We provide our customers with amended safety data sheets as prescribed by law. The customer is responsible for passing on safety data sheets and any amendments contained therein to its own customers, employees and other users of the product. We provide no guarantee that safety data sheets received by users from third parties are up-to-date. All information and instructions in this safety data sheet have been compiled to the best of our knowledge and are based on the information available to us on the day of publication. The information provided is intended to describe the product in relation to the required safety measures; it is neither an assurance of characteristics nor a guarantee of the product's suitability for particular applications and does not justify any contractual legal relationship. The existence of a safety data sheet for a particular jurisdiction does not necessarily mean

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006 - GB



Klübersynth GEM 4-150 N

Version	Revision Date:	Date of last issue: 13.11.2019	Print Date:
2.6	07.07.2020	Date of first issue: 05.11.2014	07.07.2020

that import or use within that jurisdiction is legally permitted. If you have any questions, please contact your responsible sales contact or authorized trading partner.